AGE 316
MICROCOMPUTER SYSTEMS

LECTURE 1: INTRODUCTION

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Why Are We Here?

To understand the "Brains" behind modern agricultural machinery.

To bridge the gap between **Agriculture** and **Digital Technology**.

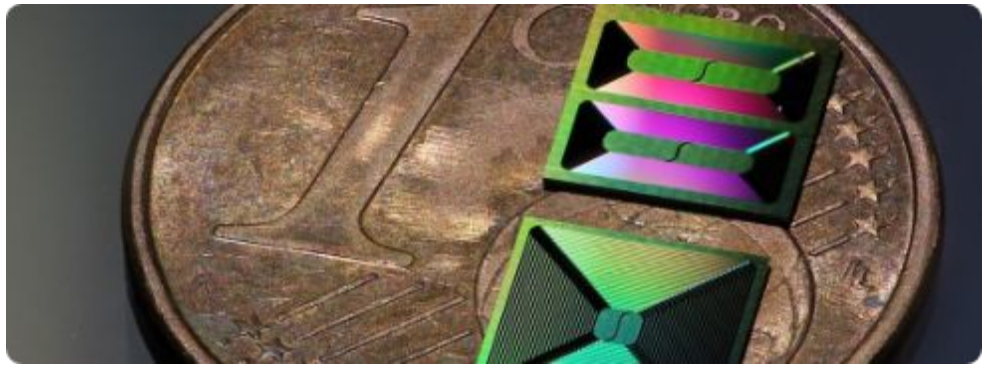
To learn how to control the physical world using code.



Processor vs. Controller

Microprocessor

"The Brain in a Jar"



- High processing power
- Needs external RAM, ROM

Microcontroller

"The Complete Organism"



- Self-sufficient System
- CPU + RAM + I/O on one chip

What is an Embedded System?

"A computing system designed for a specific task within a larger mechanical or electrical system."



Real-Time



Low Power

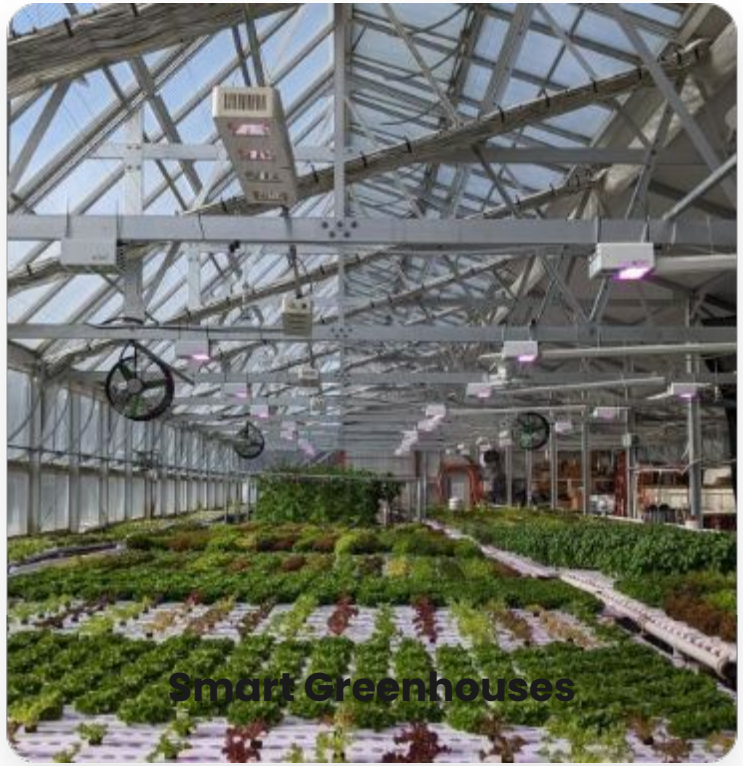


Reliable

Microcomputers in the Field



Precision Sensing

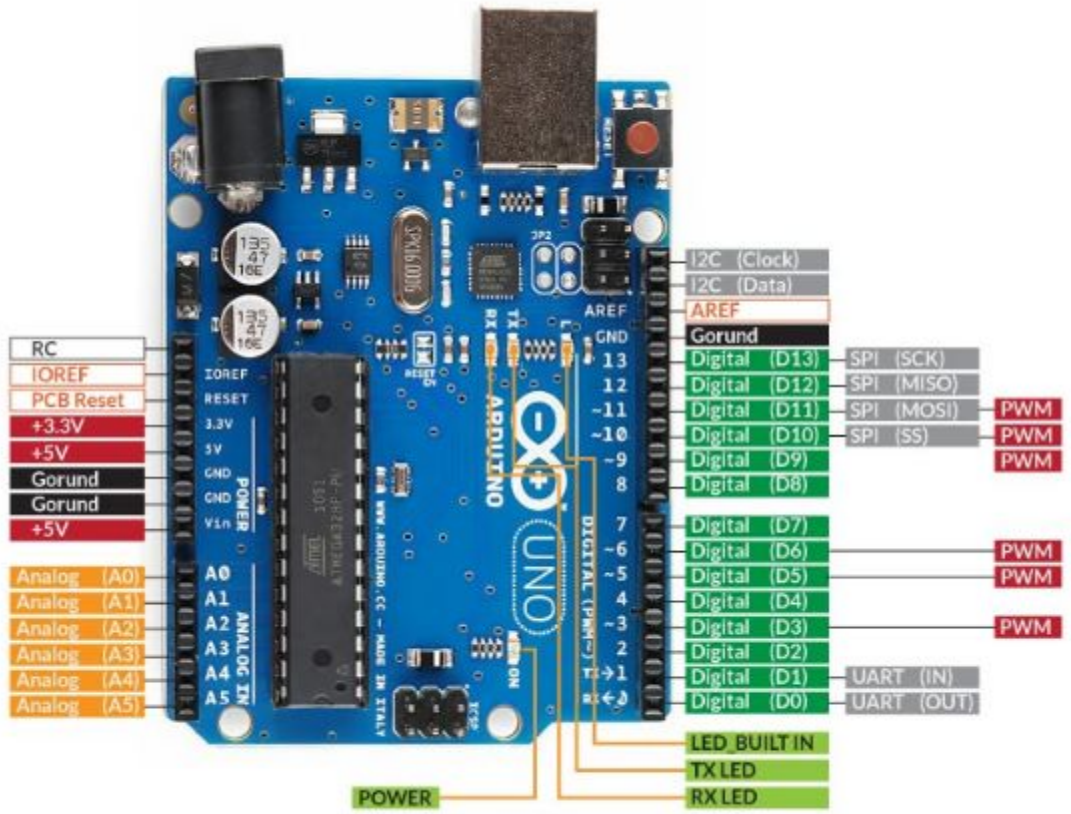


Smart Greenhouses



Agri-Robotics

Meet Your Tool: The Arduino



Arduino Uno (ATmega328P)

The world's most popular platform for learning embedded C++.

> **Why?** Open-source, affordable, and perfect for rapid agricultural prototyping.

- ✓ Digital & Analog I/O
- ✓ Easy USB Programming
- ✓ Huge Sensor Ecosystem

Course Roadmap (15 Lectures)

Hardware

Fundamentals
& Electronics

Logic

C++ Programming
for Arduino

Interfacing

Sensors, Motors
& Displays

Projects

IoT & System
Design

Preparation for Lecture 2

- [ ] **Install Arduino IDE**
  Download the latest version for your laptop.

- [ ] **Review Basic Electronics**
  Brush up on Ohm's Law (V = IR) and basic circuits.

- [ ] **Come Ready to Dissect!**
  Next class we analyze the hardware components.

Questions?

"The only stupid question is the one you don't ask."

Dr. Kadeghe Fue Sokoine
University of Agriculture

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